

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Computational Physics **L-T-P:** 2-0-0 **Credits:** 2

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Number representation: Floating-point representation; Round off error; truncation error; stability. Solution of linear algebraic equations: Gauss-Jordan elimination; LU decomposition. Random numbers: Pseudorandom numbers; Random number generation algorithms; Testing randomness (5)

Numerical differentiation and integration: Finite difference schemes to approximate derivatives (Forward and Central difference); Error assessment. Integration: Quadrature as box counting (numerical quadrature); Trapezoidal rule; Simpson's rule; Gaussian quadrature; Monte Carlo integration; Importance sampling method; von Neuman rejection method. (5)

Root finding: Bracketing and bisection method; Newton-Raphson method; Roots of polynomials; Newton-Raphson method for nonlinear systems of equations. Data fitting: Extrapolation; Linear interpolations; Lagrange interpolation; Cubic-spline interpolations; Least-Squares fitting; Linear quadratic fit; Nonlinear fit. (4)

Ordinary and partial differential equations: Euler method, Runge-Kutta (RK2, RK4, RK45) method, Adams-Bashforth Predictor-Corrector method. Eigenvalue problems using ODE solver and bisection. Two point boundary value problems: Shooting method. Elementary ideas of numerical solution of partial differential equations. (7)

Fast Fourier transform: Fourier transform of discretely sampled data, Fast Fourier transform, Convolution, correlation and autocorrelation using FFT, Power spectrum estimation using FFT. Statistical description of data: Moments of a distribution (Mean, Variance, Skewness, etc). (4)

Elementary ideas of parallel computing. Supercomputers. Examples and applications: Logistic maps; Fixed points; Period doubling; Attractors; Chaotic pendulum; Fractals; Computing fractal dimensions; Time-independent Schrödinger equation; Quantum mechanical scattering problems; Heat equation; Wave equation; Laplace equation; Poisson equation. (3)

Reference Books:

1. R. H. Landau, M.J. Páez, C. C. Bordeianu. Computational Physics: Problems Solving with Computers.
2. S. E. Koonin and D. C. Meredith. Computational Physics (Fortran Version).
3. W. H. Press, S. A. Teukolsky, W. T. Vetterling, B. P. Flanner. Numerical Recipes: The Art of Scientific Computing.

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Number representation: Floating-point representation; Round off error; truncation error;	5

	stability. Solution of linear algebraic equations: Gauss-Jordan elimination; LU decomposition. Random numbers: Pseudorandom numbers; Random number generation algorithms; Testing randomness	
2.	Numerical differentiation and integration: Finite difference schemes to approximate derivatives (Forward and Central difference); Error assessment. Integration: Quadrature as box counting (numerical quadrature); Trapezoidal rule; Simpson's rule; Gaussian quadrature; Monte Carlo integration; Importance sampling method; von Neuman rejection method.	5
3.	Root finding: Bracketing and bisection method; Newton-Raphson method; Roots of polynomials; Newton-Raphson method for nonlinear systems of equations. Data fitting: Extrapolation; Linear interpolations; Lagrange interpolation; Cubic-spline interpolations; Least-Squares fitting; Linear quadratic fit; Nonlinear fit.	4
4.	Ordinary and partial differential equations: Euler method, Runge-Kutta (RK2, RK4, RK45) method, Adams-Bashforth Predictor-Corrector method. Eigenvalue problems using ODE solver and bisection. Two point boundary value problems: Shooting method. Elementary ideas of numerical solution of partial differential equations.	7
5.	Fast Fourier transform: Fourier transform of discretely sampled data, Fast Fourier transform, Convolution, correlation and autocorrelation using FFT, Power spectrum estimation using FFT. Statistical description of data: Moments of a distribution (Mean, Variance, Skewness, etc).	4
6.	Elementary ideas of parallel computing. Supercomputers. Examples and applications: Logistic maps; Fixed points; Period doubling; Attractors; Chaotic pendulum; Fractals; Computing fractal dimensions; Time-independent Schrödinger equation; Quantum mechanical scattering problems; Heat equation; Wave equation; Laplace equation; Poisson equation .	3
Total number of hours		28